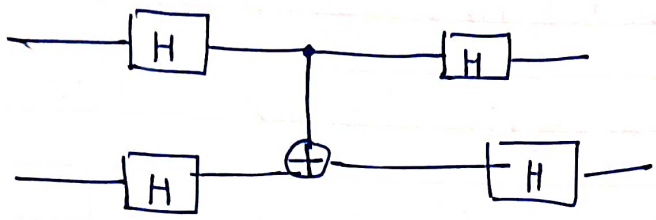
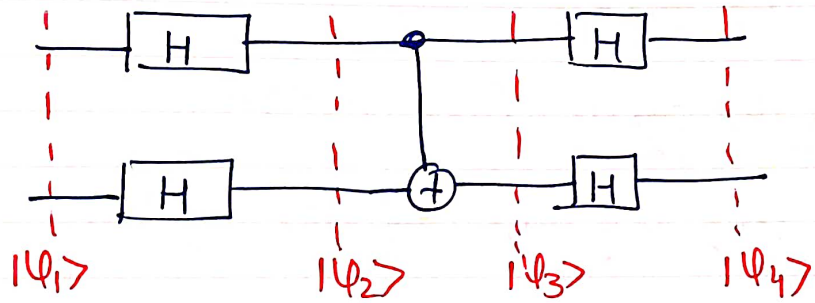
Multiple-qubit usually have multiple operations, such as in this circuit:



It is not immediately obvious what this circuit does, so let us consider the superposition that would be obtain at various times for a selection of inputs:



$|00\rangle$
$|01\rangle$
$|10\rangle$
$|11\rangle$

$|00\rangle$ Suppose first that

$|\varphi_1\rangle = |00\rangle$, then

$$|\varphi_2\rangle = \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$= \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

$$|\varphi_3\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|11\rangle + \frac{1}{2}|10\rangle$$

$$= \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$|\varphi_4\rangle = |00\rangle$$

$\because \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)H = |0\rangle$

$\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)H = 1$

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Next suppose that

$|\varphi_1\rangle = |01\rangle$, then

$$|\varphi_2\rangle = \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$= \frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$$

$$|\varphi_3\rangle = \frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle + \frac{1}{2}|11\rangle - \frac{1}{2}|10\rangle$$

$$= \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$= |11\rangle$$

Suppose $|\varphi_1\rangle = |10\rangle$, then

$$|\varphi_2\rangle = \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$= \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{1}{2}|10\rangle - \frac{1}{2}|11\rangle$$

$$|\varphi_3\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{1}{2}|11\rangle - \frac{1}{2}|10\rangle$$

$$= \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$|\varphi_3\rangle = |10\rangle$$

Finally, suppose that $|\varphi_1\rangle = |11\rangle$, then

$$|\varphi_2\rangle = \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)$$

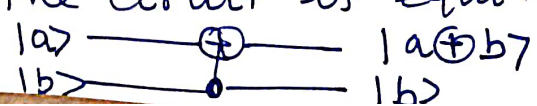
$$= \frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle - \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

$$|\varphi_3\rangle = \frac{1}{2}|00\rangle - \frac{1}{2}|01\rangle - \frac{1}{2}|11\rangle + \frac{1}{2}|10\rangle$$

$$= \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)\left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\right)$$

$$= |01\rangle.$$

It turns out that the circuit is equivalent to this:



Entanglement : Bell states:

Entangled state is a state $|\psi\rangle$ that cannot be expressed as a tensor product $|\psi_1\rangle \otimes |\psi_2\rangle$

Bell states: four 2-qubit states that are maximally entangled

$$\varphi_{00} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$\varphi_{01} = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$\varphi_{10} = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

$$\varphi_{11} = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

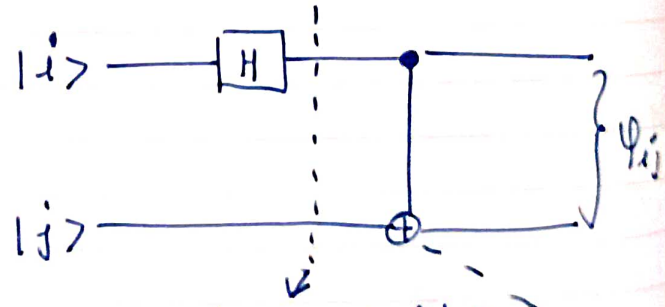
→ Measurement collapses state to $|00\rangle$ or $|11\rangle$

→ If we measure the first qubit, we automatically know the state of the second

High amount of correlation, regardless of distance.
Useful in quantum computation

Entanglement: Circuit to generate Bell states:-

Input $\lvert ij\rangle$	output $\lvert\varphi_{ij}\rangle$
$\lvert 00\rangle$	$\frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 11\rangle)$
$\lvert 01\rangle$	$\frac{1}{\sqrt{2}}(\lvert 01\rangle + \lvert 10\rangle)$
$\lvert 10\rangle$	$\frac{1}{\sqrt{2}}(\lvert 00\rangle - \lvert 11\rangle)$
$\lvert 11\rangle$	$\frac{1}{\sqrt{2}}(\lvert 01\rangle - \lvert 10\rangle)$

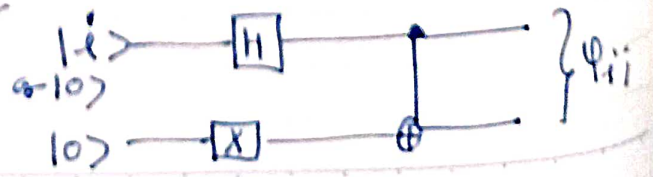


$|00\rangle$ — $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle$

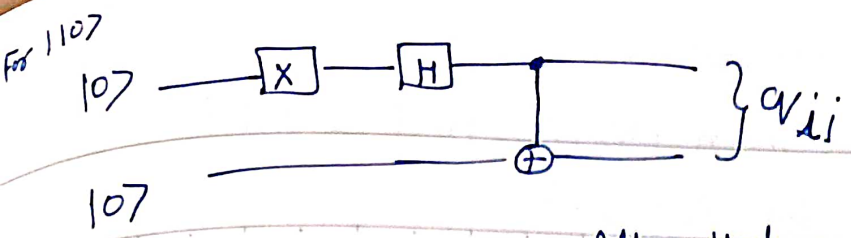
$|i\rangle = 0$ $\quad = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$

$|j\rangle = 0$

$= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$



For $|10\rangle$



$|10\rangle$ After Hadamard Gate

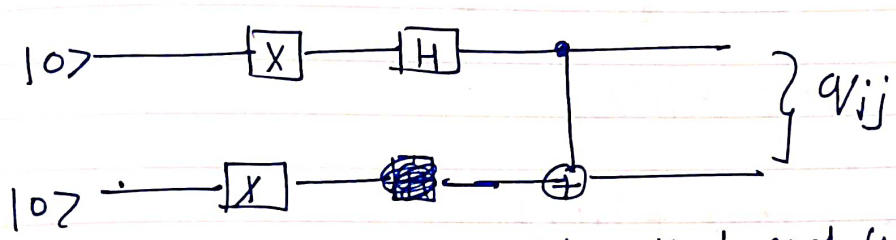
$$\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)|0\rangle$$

$$= \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle)$$

Now CNOT

$$= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

For $|11\rangle$



$|11\rangle$ After Hadamard Gate

$$= \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$$

$$\cancel{= \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)}$$

$$\cancel{= \frac{1}{2}(|00\rangle - |01\rangle - |11\rangle + |10\rangle)}$$

$$= \frac{1}{\sqrt{2}}(|01\rangle - |11\rangle)$$

After CNOT

$$= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

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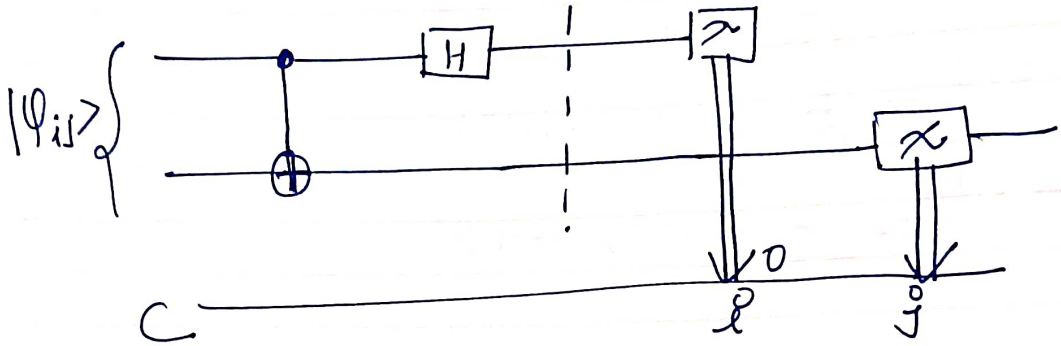
Entanglement : Bell Measurement :

The reverse problem : given $|\varphi_{ij}\rangle$ find i and j

Soln: reverse the circuit



$$|\varphi_{00}\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

After CNot $= \frac{1}{\sqrt{2}}\left(|00\rangle + |10\rangle\right)$

$$= \frac{1}{\sqrt{2}}\left(|0\rangle|0\rangle + |1\rangle|0\rangle\right)$$

then Hadamard on first gate

$$= \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)|0\rangle + \frac{1}{\sqrt{2}}\left(|0\rangle - |1\rangle\right)|0\rangle\right)$$

$$= \frac{1}{2}\left(|00\rangle + |10\rangle + |00\rangle - |10\rangle\right)$$

$$= |00\rangle$$

Input (Bell State) $|\varphi_{ij}\rangle$	output ij
$\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$	0 0
$\frac{1}{\sqrt{2}}(\|01\rangle + \|10\rangle)$	0 1
$\frac{1}{\sqrt{2}}(\|00\rangle - \|11\rangle)$	1 0
$\frac{1}{\sqrt{2}}(\|01\rangle - \|10\rangle)$	1 1

No.

Date

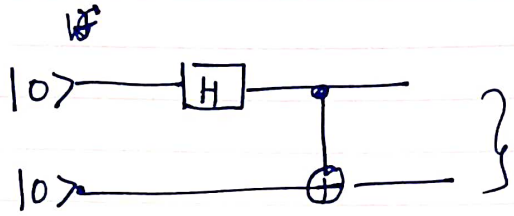
Bell State:-

What are Bell States?
How many Bell States exist?
How to create a Bell State?

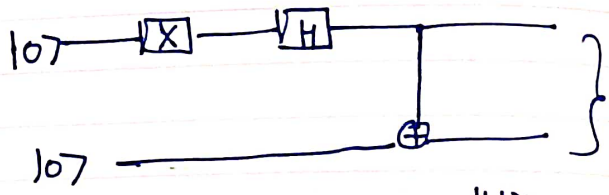
In quantum information science, the Bell's states or EPR pairs are specific quantum states of two qubits that represent the simplest examples of quantum entanglement. The Bell's states are a form of entangled and normalized basis vectors.

"The Bell states are four specific maximally entangled quantum states of two qubits".

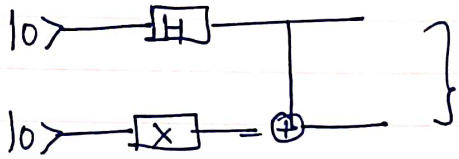
00
10
01
11



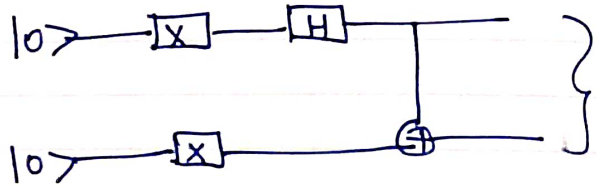
$$|\varphi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$



$$|\varphi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

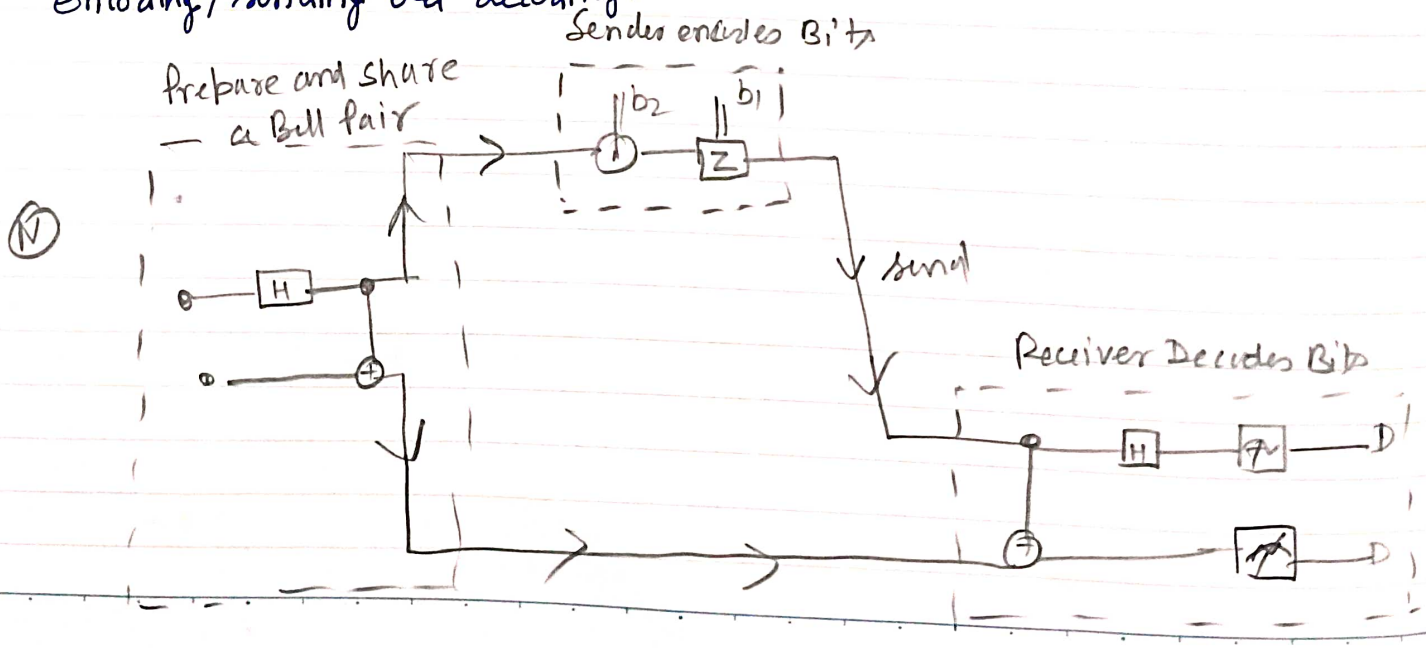


$$|\phi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

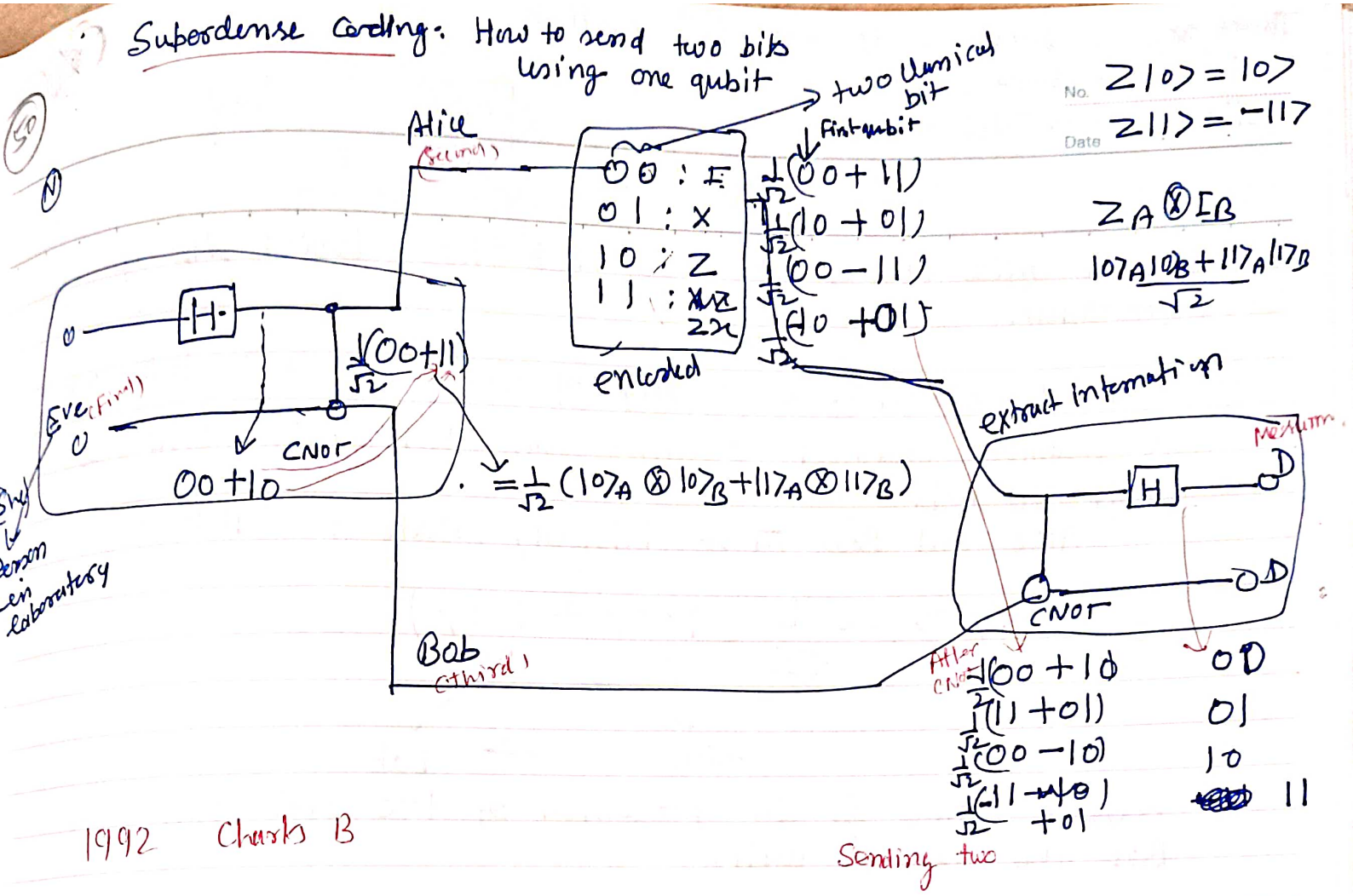


$$|\phi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

The protocol can be split into five different steps: preparation, sharing, encoding, sending and decoding.



Superdense Coding: How to send two bits using one qubit



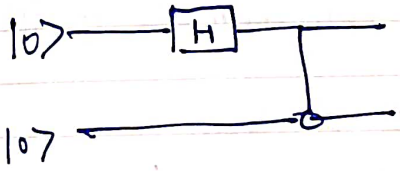
$Z|0\rangle = |0\rangle$

$Z|1\rangle = -|1\rangle$

$Z_A \otimes I_B$

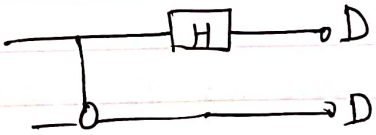
$\frac{|0\rangle_A|0\rangle_B + |1\rangle_A|1\rangle_B}{\sqrt{2}}$

1992 Charles B



$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

Bell State



$X_A \otimes I_B \left(\frac{|0\rangle_A|0\rangle_B + |1\rangle_A|1\rangle_B}{\sqrt{2}} \right)$

X on first
I on second

$\frac{X_A|0\rangle_A \otimes I_B|0\rangle_B + X_A|1\rangle_A \otimes I_B|1\rangle_B}{\sqrt{2}}$

$= \frac{|0\rangle + |1\rangle}{\sqrt{2}}$

There are two protocols to send information:

(i) Super-dense coding

(ii) Quantum Teleportation

Alice "She" | Bob

Super-dense coding :- (How to send two bits using one qubit)

Suppose Alice wishes to send Bob two classical bits of information.

↓

Superdense coding is a way of achieving this task over a quantum channel, requiring only that Alice send one qubit to Bob.

Alice and Bob must initially share the Bell state

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Alice ⟶ Bob

Suppose Alice is in procession of the first qubit and Bob the second qubit.

Alice performs one of four 1-qubit gates, depending on the 2 classical bits she wishes to communicate to Bob.

↓

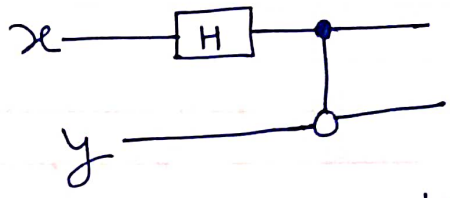
- If Alice wishes to send the bits 00 to Bob, she does nothing to her qubit (or equivalently, applies the identity gate I).
- If She wishes to send 01, she applies the X-gate to her qubit.
- If she wishes to send 10, she applies the Z-gate to her qubit.
- If she wishes to send 11, she applies Z.X (that is she applies the X gate followed by the Z gate).

(N) $I \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $X \equiv \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $Y \equiv \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ $Z \equiv \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

To send	Transformation	
00	$I \otimes I$:	$\frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 11\rangle) \Leftrightarrow \frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 11\rangle) = \lvert\beta_{00}\rangle$
01	$X \otimes I$:	$\frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 11\rangle) \Leftrightarrow \frac{1}{\sqrt{2}}(\lvert 10\rangle + \lvert 01\rangle) = \lvert\beta_{01}\rangle$
10	$Z \otimes I$:	$\frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 11\rangle) \Leftrightarrow \frac{1}{\sqrt{2}}(\lvert 00\rangle - \lvert 11\rangle) = \lvert\beta_{10}\rangle$
11	$Z \cdot X \otimes I$:	$\frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 11\rangle) \Leftrightarrow \frac{1}{\sqrt{2}}(-\lvert 10\rangle + \lvert 01\rangle) = \lvert\beta_{11}\rangle$

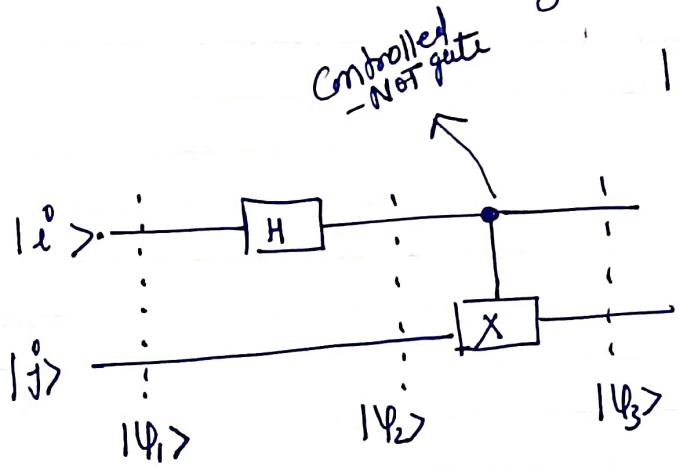
(This basis is known as the Bell basis, and the four states $\lvert\beta_{00}\rangle, \lvert\beta_{01}\rangle, \lvert\beta_{10}\rangle, \lvert\beta_{11}\rangle$ are called the Bell states (also called EPR pairs).

$$\alpha\lvert 0\rangle + \beta\lvert 1\rangle \quad \text{—}\boxed{H}\text{—} \quad \alpha\frac{\lvert 0\rangle + \lvert 1\rangle}{\sqrt{2}} + \beta\frac{\lvert 0\rangle - \lvert 1\rangle}{\sqrt{2}}$$



$\lvert\beta_{xy}\rangle$

$$\lvert\beta_{xy}\rangle = \frac{\lvert 0, y\rangle + (-1)^x \lvert 1, \bar{y}\rangle}{\sqrt{2}}$$



$\lvert\beta_{ij}\rangle$

$\lvert\varphi_1\rangle = \lvert 00\rangle$

$\lvert\varphi_2\rangle = H\lvert i\rangle$

$= \frac{1}{\sqrt{2}}(\lvert 0\rangle + \lvert 1\rangle)\lvert 0\rangle$

$= \frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 10\rangle)$

$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$H\lvert 0\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Next, the Controlled-NOT gate transforms this state into

$\lvert\varphi_3\rangle = \frac{1}{\sqrt{2}}(\lvert 00\rangle + \lvert 11\rangle)$

$= \lvert\beta_{00}\rangle$

Alice sends her qubit to Bob. Then Bob is in possession of one of the four Bell states, depending on the classical bits. Alice wished to send him.

↓

Bob can now simply perform a measurement of the joint 2-qubit state with respect to the Bell basis (i.e. the basis $\{|B_{00}\rangle, |B_{01}\rangle, |B_{10}\rangle, |B_{11}\rangle\}$).

↓

Such a measurement can be implemented by first performing a change of basis to the Bell basis, and then performing a measurement in the computational basis.

↓

The outcome of the Bell measurement reveals to Bob which Bell state he possesses, and so allows him to determine with certainty the two classical bits Alice wanted to communicate to him.

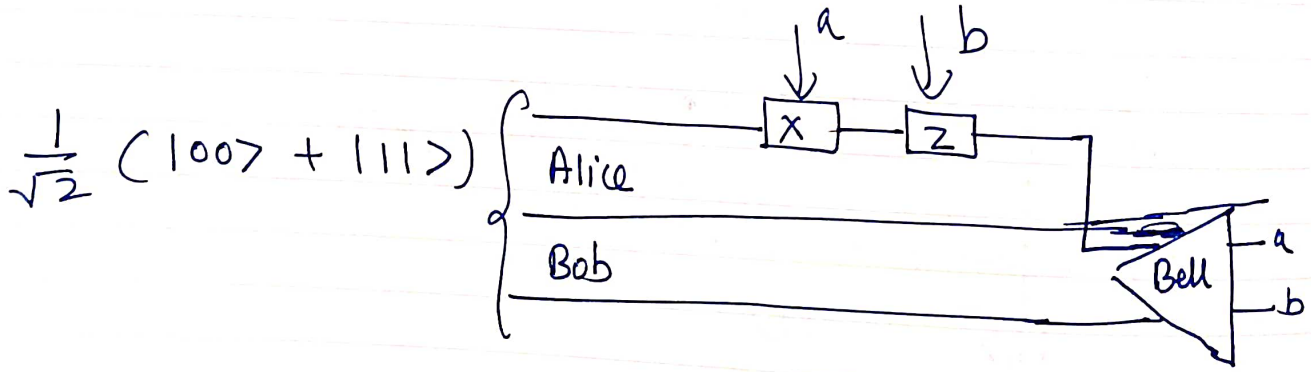


Fig:. The superdense coding protocol in which Alice sends two bits of classical information by sending one physical qubit to Bob. Alice and Bob initially share an EPR pair $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Alice applies the operation $U_{ab} = Z^b X^a$ depending on the classical bits a, b that she wishes to send. After sending her qubit to Bob, he measures his pair of qubits in the Bell basis. This measurement gives Bob the two values (a) and (b) corresponding to the Bell state $|B_{ab}\rangle$ in his procession.

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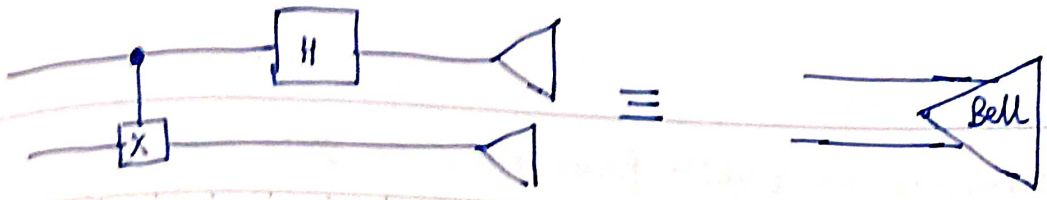
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